Notebook 6 — Chapter 6.1

Dynamic Programming on Random Variables

20 worked examples — finite-horizon stopping, reservation values, and stopping with per-step costs — calibrated for Stage 1/Stage 2 speed and rigor

The Master Recursion (read this first)

You may draw X (a die roll, an offer, a price) up to k times. After each draw you either *take* it or *pass* and draw again. Let V_j be your expected payoff at the *start* of draw j under optimal play. On the last draw you must take what you get:

$$V_k = \mathbb{E}[X]$$

Working backwards, at draw j the value of passing is V_{j+1}, so you take X exactly when it beats that:

$$V_j = \mathbb{E}\big[\max(X,\, V_{j+1})\big] \qquad \text{accept } X \text{ iff } X > V_{j+1}$$

V_{j+1} is your *reservation value* at draw j. With a cost c per extra draw, passing is worth $V_{j+1} - c$ instead.
Evaluating $\mathbb{E}[\max(X,t)]$ **fast:** faces below t get *replaced by* t; faces above t stay.

A. Backward Induction on Dice

Example 1 — One Re-roll

You roll a fair 6-sided die and may either keep the result or re-roll once (keeping the second result). Find the expected payoff and the optimal rule.
Solution. Here $k = 2$. The last roll is forced: $V_2 = \mathbb{E}[X] = \frac{7}{2}$.
At roll 1, passing is worth $V_2 = 3.5$, so keep X iff $X > 3.5$, i.e. $X \in \{4,5,6\}$. Faces $1,2,3$ are replaced by 3.5:

$$V_1 = \mathbb{E}[\max(X,3.5)] = \frac{3.5+3.5+3.5+4+5+6}{6}$$

$$= \frac{10.5+15}{6} = \boxed{\tfrac{17}{4} = 4.25}$$

⋆ *Analogy:* the re-roll is a *safety net* worth exactly 3.5. You only keep a roll that beats the net — the net lifts every bad outcome up to 3.5 and leaves the good ones alone.

Example 2 — Three Rolls

Same die, but you may roll up to 3 times. Find V_1 and the accept rule at each roll.
Solution. Backward: $V_3 = \frac{7}{2}$ and $V_2 = \frac{17}{4}$ (Example 1).
At roll 1 the net is $V_2 = 4.25$, so keep $X \in \{5,6\}$; faces 1–4 become 4.25:

$$V_1 = \frac{4(4.25)+5+6}{6} = \frac{17+11}{6} = \boxed{\tfrac{14}{3} \approx 4.667}$$

Rule: roll 1 accept ≥ 5; roll 2 accept ≥ 4 (net $V_3 = 3.5$); roll 3 take anything.
⋆ *Analogy:* a *patience staircase*. Early on you can afford to be picky because plenty of tries remain; each step down the staircase you lower your standards as the safety net shrinks.

Example 3 — The Value Ladder and Diminishing Returns

Compute V_1 for $k = 1,2,3,4,5$ rolls of a fair die and the *marginal value* of each extra re-roll.
Solution. Each step feeds the previous value into $\mathbb{E}[\max(X,\cdot)]$:

$$k=4: \quad V_1 = \frac{4\cdot\frac{14}{3}+5+6}{6} = \frac{89}{18} \approx 4.944$$

$$k=5: \quad V_1 = \frac{4\cdot\frac{89}{18}+5+6}{6} = \frac{277}{54} \approx 5.130$$

k	1	2	3	4	5
V_1	3.5	4.25	4.667	4.944	5.130
Δ		0.75	0.417	0.278	0.185

Gains shrink fast, and $V_1 \to 6$ only slowly.
▶ *Speed trick:* for a fair d-sided die and net t with $m = \lfloor t \rfloor$ rejected faces,

$$\mathbb{E}[\max(X,t)] = \frac{m\,t + \left[\frac{d(d+1)}{2} - \frac{m(m+1)}{2}\right]}{d}$$

No need to write all faces out.

Example 4 — A 20-Sided Die

A fair 20-sided die may be rolled up to 2 times, and also up to 3 times. Find V_1 in each case.
Solution. $k = 2$: $V_2 = 10.5$, so keep $X \geq 11$. The kept faces sum to $11 + \cdots + 20 = 155$:

$$V_1 = \frac{10\cdot 10.5 + 155}{20} = \frac{105+155}{20} = \boxed{13}$$

$k = 3$: the net at roll 1 is 13, so faces 1–13 (thirteen of them) become 13 and faces 14–20 stay; their sum is 119:

$$V_1 = \frac{13\cdot 13 + 119}{20} = \frac{288}{20} = \boxed{\tfrac{72}{5} = 14.4}$$

Accept rule at roll 1 with three rolls: $X \geq 14$.
⋆ *Analogy:* the second roll lifts a d20 from 10.5 to 13 — a gain of 2.5, while on a d6 it gave 0.75. The option is worth more on a wider die because there is more upside to chase.

Example 5 — How Long Does the Game Last?

In the optimal 3-roll d6 game of Example 2, find $\mathbb{E}[\#\text{rolls}]$ and the distribution of the final payoff.
Solution. Stop at roll 1 iff $X \geq 5$: probability $\frac{1}{3}$. Reach roll 2 with probability $\frac{2}{3}$; stop there iff $X \geq 4$: probability $\frac{1}{2}$. Reach roll 3 with probability $\frac{2}{3}\cdot\frac{1}{2} = \frac{1}{3}$.

$$\mathbb{E}[\#\text{rolls}] = 1 + \tfrac{2}{3} + \tfrac{1}{3} = \boxed{2}$$

Final payoff $= 6$: take a 6 on roll 1, 2 or 3:

$$\tfrac{1}{6} + \tfrac{2}{3}\cdot\tfrac{1}{6} + \tfrac{1}{3}\cdot\tfrac{1}{6} = \tfrac{1}{3}$$

Similarly $\mathbb{P}(5) = \frac{1}{3}$, $\mathbb{P}(4) = \frac{1}{6}$ (roll 2 or 3 only), and $\mathbb{P}(1) = \mathbb{P}(2) = \mathbb{P}(3) = \frac{1}{18}$ (roll 3 only).
Check: $\mathbb{E} = \frac{11}{3} + \frac{4}{6} + \frac{6}{18} = \frac{14}{3}$ ✓.
⋆ *Analogy:* a hiring funnel — you interview about 2 candidates on average, and the final hire is usually a top candidate, yet one time in six you are stuck with whoever is left in the final round.

B. Reservation Values on Other Distributions

Example 6 — A Skewed Die: the Threshold Is the Mean

A fair die has faces $\{2,3,5,8,13,21\}$ and may be rolled up to 2 times. Find V_1 and the accept rule.
Solution. $V_2 = \mathbb{E}[X] = \frac{52}{6} = \frac{26}{3} \approx 8.667$. Keep a face iff it exceeds 8.667: only 13 and 21. The faces $2,3,5,8$ are replaced by $\frac{26}{3}$:

$$V_1 = \frac{4\cdot\frac{26}{3} + 13 + 21}{6} = \frac{\frac{104}{3}+34}{6} = \boxed{\tfrac{103}{9} \approx 11.44}$$

⋆ *Analogy:* the safety net is the *mean of the whole die*, not the middle face. Here an 8 looks like a "middle" outcome but is thrown back — the average is dragged up by the two big faces.

Example 7 — Uniform Offers

Offers $U \sim \text{Uniform}(0,1)$ arrive one at a time, up to 3 offers; you may accept an offer or pass. Find V_1 and the thresholds.
Solution. For a net $t \in [0,1]$, offers below t become t (probability t) and offers above stay:

$$\mathbb{E}[\max(U,t)] = t\cdot t + \int_t^1 u\,du = t^2 + \tfrac{1-t^2}{2} = \frac{1+t^2}{2}$$

$$V_3 = \tfrac{1}{2}, \qquad V_2 = \frac{1+\frac{1}{4}}{2} = \tfrac{5}{8}$$

$$V_1 = \frac{1+\frac{25}{64}}{2} = \boxed{\tfrac{89}{128} \approx 0.695}$$

Rule: offer 1 accept if $> \frac{5}{8}$; offer 2 accept if $> \frac{1}{2}$; offer 3 forced.
▶ *Speed trick:* the Uniform$(0,1)$ recursion is simply $V_{\text{new}} = \frac{1+V_{\text{old}}^2}{2}$ — one squaring per extra draw.

Example 8 — Exponential Offers and the Memoryless Shortcut

Offers $X \sim \text{Exp}(1)$, up to 3 offers. Find V_1.
Solution. Write $\max(X,t) = t + (X-t)^+$. By memorylessness, given

$X > t$ the excess $X - t$ is again Exp(1), and $\mathbb{P}(X > t) = e^{-t}$:

$$\mathbb{E}[\max(X, t)] = t + e^{-t}$$

$$V_3 = 1, \qquad V_2 = 1 + e^{-1} \approx 1.368$$

$$V_1 = V_2 + e^{-V_2} \approx 1.368 + 0.255 = \boxed{1.623}$$

Rule: offer 1 accept if > 1.368; offer 2 accept if > 1.
⋆ *Analogy:* with a heavy tail the "jackpot" is open-ended. Three tries lift the payoff from 1 to 1.62 — a 62% gain — versus only +33% for three rolls of a bounded d6.

Example 9 — The Cost of the Wrong Threshold

In the 3-roll d6 game, a player accepts *only a 6* on rolls 1 and 2 (and takes whatever comes on roll 3). Find her value and compare with Example 2.
Solution. Work backwards. Roll 3 forced: 3.5. At roll 2: take a 6 w.p. $\frac{1}{6}$, else pass into 3.5:

$$W_2 = \tfrac{1}{6} \cdot 6 + \tfrac{5}{6} \cdot \tfrac{7}{2} = 1 + \tfrac{35}{12} = \tfrac{47}{12}$$

$$W_1 = \tfrac{1}{6} \cdot 6 + \tfrac{5}{6} \cdot \tfrac{47}{12} = 1 + \tfrac{235}{72} = \boxed{\tfrac{307}{72} \approx 4.264}$$

Optimal was $\frac{14}{3} \approx 4.667$: the over-picky rule loses ≈ 0.40 per game.
▶ *Speed trick:* a stopping rule *always* has a computable value — evaluate it by the same backward recursion. Comparing a heuristic to the DP optimum shows exactly what your intuition costs.

Example 10 — Does the Option Also Cut Risk?

For the 2-roll d6 game (Example 1), find Var(final payoff) and compare with a single roll.
Solution. Roll 1 is kept if ≥ 4. Each of $4, 5, 6$ ends the game at roll 1 (probability $\frac{1}{6}$ each), or arrives on roll 2 w.p. $\frac{1}{2} \cdot \frac{1}{6}$:

$$\mathbb{P}(4) = \mathbb{P}(5) = \mathbb{P}(6) = \tfrac{1}{4}, \qquad \mathbb{P}(1) = \mathbb{P}(2) = \mathbb{P}(3) = \tfrac{1}{12}$$

$$\mathbb{E}[X^2] = \tfrac{16+25+36}{4} + \tfrac{1+4+9}{12} = \tfrac{245}{12}$$

$$\mathrm{Var} = \tfrac{245}{12} - \left(\tfrac{17}{4}\right)^2 = \boxed{\tfrac{113}{48} \approx 2.354}$$

A single roll has variance $\frac{35}{12} \approx 2.917$.
⋆ *Analogy:* the safety net cuts off the left tail. You raise the mean *and* shrink the spread — a rare case where more return comes with less risk.

C. Stopping With Costs

Example 11 — Unlimited Rolls, Each Costs $\frac{1}{2}$

You may roll a fair die as many times as you like; *every* roll (including the first) costs $c = \frac{1}{2}$, and your payoff is the final roll minus all costs paid. Find the optimal rule and the value.
Solution. Let C be the value of "pay for one more roll and continue optimally". Stationarity gives

$$C = -c + \mathbb{E}[\max(X, C)]$$

Use $\max(X, C) = C + (X - C)^+$ to rewrite it as a clean equation:

$$\boxed{\mathbb{E}\big[(X - C)^+\big] = c}$$

Try $C = 4$: $\mathbb{E}[(X-4)^+] = \frac{1+2}{6} = \frac{1}{2} = c$ ✓. So $C = 4$.
Rule: stop on $X > 4$, i.e. on 5 or 6 (a 4 is a tie). *Check:* $\mathbb{E}[\#\text{rolls}] = 1/\frac{1}{3} = 3$, so

$$\text{value} = \mathbb{E}[X \mid X \geq 5] - c \cdot 3 = 5.5 - 1.5 = \boxed{4}$$

⋆ *Analogy:* the reservation value is where *expected extra gain from rolling equals its ticket price.* Pay $\frac{1}{2}$ per try and you aim for ≥ 5; the average winning hand is 5.5 but it takes 3 tickets to win.

Example 12 — How the Cost Moves the Threshold

Same game as Example 11 with (a) $c = 1$, (b) $c = \frac{1}{4}$, (c) $c = \frac{5}{2}$. Find the continuation value C and the rule.
Solution. Solve $\mathbb{E}[(X - C)^+] = c$ piece by piece.
(a) $C = 3$: $\frac{1+2+3}{6} = 1$ ✓. Stop on ≥ 4.
(b) For $C \in [4, 5]$: $\frac{(5-C)+(6-C)}{6} = \frac{1}{4} \Rightarrow 11 - 2C = \frac{3}{2} \Rightarrow C = \frac{19}{4} = 4.75$. Stop on ≥ 5.
(c) For $C \leq 1$: every face exceeds C, so $\mathbb{E}[(X - C)^+] = 3.5 - C = \frac{5}{2} \Rightarrow C = 1$. Since $X \geq 1$, never re-roll: **stop on the first roll.**

c	1	$\frac{1}{4}$	$\frac{5}{2}$
C	3	4.75	1

▶ *Speed trick:* when $c \geq \mathbb{E}[X] - \min X$ the option is worthless: re-rolling can never repay its cost, so take the first offer.

Example 13 — Finite Horizon With a Re-roll Fee

Up to 3 rolls of a d6; the first roll is free, and *each re-roll costs* $\frac{1}{2}$. Find V_1 and the rules.
Solution. Passing at roll j is now worth $V_{j+1} - c$:

$$V_j = \mathbb{E}\big[\max(X,\, V_{j+1} - c)\big]$$

$V_3 = \frac{7}{2}$. At roll 2 the net is $3.5 - 0.5 = 3$: faces $1, 2, 3 \to 3$:

$$V_2 = \frac{3 \cdot 3 + 4 + 5 + 6}{6} = \frac{24}{6} = 4$$

At roll 1 the net is $4 - 0.5 = 3.5$:

$$V_1 = \mathbb{E}[\max(X, 3.5)] = \boxed{\tfrac{17}{4}}$$

Rules: roll 1 accept ≥ 4; roll 2 accept ≥ 4 (a 3 is a tie).
⋆ *Analogy:* three rolls with a $\frac{1}{2}$ fee are worth exactly the same as *two free rolls* (Example 1) — the second re-roll's value is eaten entirely by its fee.

Example 14 — An Early-Exit Penalty

Up to 3 rolls of a d6. If you stop at roll 1 or 2 you pay an early-exit fee of $\frac{1}{2}$ (payoff $X - \frac{1}{2}$); stopping at the last roll is free. Find V_1.
Solution. Now *taking* the roll is what costs, so compare $X - \frac{1}{2}$ with the continuation:

$$V_j = \mathbb{E}\big[\max(X - \tfrac{1}{2},\, V_{j+1})\big] \quad (j < 3), \qquad V_3 = \tfrac{7}{2}$$

Roll 2: $X - \frac{1}{2} > 3.5 \iff X \geq 5$ (a 4 ties). Faces 1–4 give 3.5:

$$V_2 = \frac{4(3.5) + 4.5 + 5.5}{6} = 4$$

Roll 1: $X - \frac{1}{2} > 4 \iff X \geq 5$:

$$V_1 = \frac{4(4) + 4.5 + 5.5}{6} = \boxed{\tfrac{13}{3} \approx 4.333}$$

⋆ *Analogy:* a cancellation fee on a hotel booking. It makes you more reluctant to *commit* early, so you demand a higher raw roll before paying to exit — the value sits between 2 and 3 free rolls ($4.25 < 4.33 < 4.67$).

Example 15 — Job Search With Uniform Offers

Offers are Uniform$(0, 1)$; each draw costs $c = 0.02$ (including the first); you may draw forever. Find the reservation value, $\mathbb{E}[\#\text{draws}]$ and the net value.
Solution. Use $\mathbb{E}[(U - C)^+] = c$:

$$\int_C^1 (u - C)\, du = \frac{(1 - C)^2}{2} = c \;\Rightarrow\; C = 1 - \sqrt{2c} = 1 - 0.2 = \boxed{0.8}$$

Accept any offer > 0.8: $\mathbb{P}(\text{accept}) = 0.2$, so draws are geometric with $\mathbb{E}[\#] = \frac{1}{0.2} = 5$. The accepted offer averages 0.9:

$$\text{net} = 0.9 - 5(0.02) = 0.8 = C \;\checkmark$$

⋆ *Analogy:* the classic *McCall job search.* You hold out for an offer above 80% of the best possible; a lower price of search (c) raises your standards as $1 - \sqrt{2c}$.

D. Capstones

Example 16 — Exponential Search: Log Scaling

Offers are Exp(1), each draw costs $c = 0.1$, unlimited draws. Find the reservation value, $\mathbb{E}[\#\text{draws}]$ and the net value.
Solution. By memorylessness $\mathbb{E}[(X-C)^+] = e^{-C}$, so

$$e^{-C} = c \;\Rightarrow\; C = \ln\tfrac{1}{c} = \ln 10 \approx \boxed{2.303}$$

Accept $X > C$: $\mathbb{P} = e^{-C} = 0.1$, so $\mathbb{E}[\#] = 10$ draws. The accepted offer averages $C + 1 = 3.303$ (memorylessness again):

$$\text{net} = 3.303 - 10(0.1) = 2.303 = C \;\checkmark$$

⋆ *Analogy:* to raise your *standard* by 1 you must make search e times cheaper. The reservation value grows only with $\ln(1/c)$ — a tenfold cut in costs adds just $\ln 10 \approx 2.3$ to your threshold.

Example 17 — Discounting the Future

A re-roll arrives one period later and its payoff is multiplied by $\delta = 0.9$. (a) With unlimited rolls of a d6, find V and the rule. (b) With at most 3 rolls, find V_1.
Solution. (a) $V = \mathbb{E}[\max(X, \delta V)]$. Guess $\delta V \in [4, 5)$, so only $5, 6$ are kept:

$$V = \frac{4\delta V + 11}{6} \;\Rightarrow\; V = \frac{11}{6 - 4\delta} = \frac{11}{2.4} = \boxed{\tfrac{55}{12} \approx 4.583}$$

Check: $\delta V = 4.125 \in [4, 5)$ ✓, so the guess is self-consistent (accept ≥ 5).
(b) $V_3 = 3.5$; roll 2 net $0.9(3.5) = 3.15$:

$$V_2 = \frac{3(3.15) + 15}{6} = 4.075$$

Roll 1 net $0.9(4.075) = 3.6675$ (keep ≥ 4):

$$V_1 = \frac{3(3.6675) + 15}{6} = \boxed{4.33375}$$

▶ *Speed trick:* when the optimal rule is *stationary*, guess the accept set, solve the linear equation for V, then verify the guess is consistent — far faster than iterating.

Example 18 — How Many Uniform Draws to Reach 0.9?

Using $V_{\text{new}} = \frac{1+V_{\text{old}}^2}{2}$ for Uniform$(0, 1)$ offers, find the smallest number of draws with $V_1 \geq 0.9$, and describe the long-run behaviour.
Solution. Iterate from $V = \frac{1}{2}$:

n	1	2	3	4	5	8	12
V_1	.500	.625	.695	.742	.775	.836	.879

Continuing, $n = 15$ gives $0.8986 < 0.9$ and $n = 16$ gives $\boxed{0.9037}$: **16 draws**.
Long run: let $e_n = 1 - V_1$. Then $1 - \frac{1+V^2}{2} = \frac{(1-V)(1+V)}{2} = e - \frac{e^2}{2}$, so

$$e_{n+1} = e_n - \tfrac{e_n^2}{2} \;\Rightarrow\; \frac{d}{dn}\left(\frac{1}{e}\right) \approx \frac{1}{2} \;\Rightarrow\; e_n \approx \frac{2}{n + \text{const}}$$

The shortfall from the maximum decays only like $2/n$.
⋆ *Analogy:* each extra try removes only a sliver proportional to the *square* of the shortfall still left, so progress slows as you approach the ceiling. Going from 0.9 to 0.99 needs about 10× more draws.

Example 19 — A Patience Ladder With Costs

Offers Uniform$(0, 1)$, at most 4 draws, each re-draw costs $c = 0.02$ (first draw free). Find V_1 and the thresholds.
Solution. Passing is worth $V_{j+1} - c$, so with $\mathbb{E}[\max(U, t)] = \frac{1+t^2}{2}$:

$$V_4 = 0.5$$

$$V_3 = \tfrac{1+(0.5-0.02)^2}{2} = \tfrac{1+0.2304}{2} = 0.6152$$

$$V_2 = \tfrac{1+(0.6152-0.02)^2}{2} = \tfrac{1+0.3543}{2} = 0.6771$$

$$V_1 = \tfrac{1+(0.6771-0.02)^2}{2} = \boxed{0.7159}$$

Thresholds (the net at each draw): draw 1 accept > 0.657, draw 2 accept > 0.595, draw 3 accept > 0.48, draw 4 forced.
⋆ *Analogy:* a limit order working a shrinking time budget — you bid high at first, then concede a little at each rung of the ladder as the clock runs down.

Example 20 — Capstone: 4 Rolls With a Fee

A d6 may be rolled up to 4 times; the first roll is free and each re-roll costs $\frac{1}{4}$. Find V_1, the accept rule at every roll, and the most you would pay to play this game instead of a single roll.
Solution. $V_4 = \frac{7}{2}$. Passing at roll j is worth $V_{j+1} - \frac{1}{4}$.
Roll 3: net 3.25 (faces 1–3 lifted): $V_3 = \frac{3(3.25)+15}{6} = \frac{33}{8}$.
Roll 2: net $\frac{33}{8} - \frac{1}{4} = \frac{31}{8} = 3.875$: $V_2 = \frac{3(3.875)+15}{6} = \frac{71}{16}$.
Roll 1: net $\frac{71}{16} - \frac{1}{4} = \frac{67}{16} = 4.1875$ (faces 1–4 lifted):

$$V_1 = \frac{4 \cdot \frac{67}{16} + 11}{6} = \boxed{\tfrac{37}{8} = 4.625}$$

Rules: roll 1 accept ≥ 5; roll 2 accept ≥ 4; roll 3 accept ≥ 4; roll 4 forced.
Entry price: versus one roll (3.5), the game is worth

$$\tfrac{37}{8} - \tfrac{7}{2} = \boxed{\tfrac{9}{8} = 1.125}$$

Without the fee, four rolls would be worth $\frac{89}{18} \approx 4.944$, so the fees consume ≈ 0.32.
▶ *Full-circle intuition:* the same backward recursion $V_j = \mathbb{E}[\max(X, V_{j+1} - c)]$ prices every game in this chapter — change the distribution, the horizon, the fee or the discount, and only the net $V_{j+1} - c$ changes. Build the ladder from the last step upward.